SGA 2: Application to the Fundamental Group

Lemma 3.9 — proof

Bounded source-aligned working unit — 22 July 2026

Authority and boundary. Corrected French TeX lines 3555–3566. The unit begins on original printed p. 120 and continues on printed p. 121. Its first portion is on source-PDF physical p. 104 (recomposed running p. 96); the final sentence continues on physical p. 105 (running p. 97). Blank line 3567 is excluded. The raw continuation cursor is line 3567 and the next substantive cursor is line 3568. The French authority remains byte-identical. The same-edition PDF is manifestation/layout evidence only, and the external English candidate is comparison only. Independent review and archive handoff have not been performed; this package is internal and not for release.

Let $X' = \operatorname{Spec}(A)$ and let $Y' = V(t)$, which we identify with the spectrum of B. Let $x = \mathfrak{r}(A)$, and set $X = X' \setminus \{x\}$ and $Y = Y' \setminus \{x\} = X \cap Y'$. Denote by $\widehat{X'}$ the formal spectrum of A for the t-adic topology; it is identified with the formal completion of X' along Y'.

Since A is complete for the t-adic topology, observe that $\mathbf{Et}(X') \longrightarrow \mathbf{Et}(\widehat{X'})$ is an equivalence of categories. Likewise, $\mathbf{Et}(\widehat{X'}) \longrightarrow \mathbf{Et}(Y')$ is an equivalence by Proposition 1.1; hence $\mathbf{Et}(X') \longrightarrow \mathbf{Et}(Y')$ is an equivalence of categories.

We prove (i). Consider the diagrams

$$\begin{tikzcd} X' & X \arrow[l] \\ Y' \arrow[u] & Y \arrow[l] \arrow[u] \end{tikzcd} \qquad \begin{tikzcd} \mathbf{Et}(X') \arrow[r, "a"] \arrow[d, "c"'] & \mathbf{Et}(X) \arrow[d, "b"] \\ \mathbf{Et}(Y') \arrow[r, "d"] & \mathbf{Et}(Y) \end{tikzcd}$$

We have just seen that c is an equivalence; d is also an equivalence by the hypothesis that B is pure; and finally b is fully faithful, as was seen in Example 2.1; cf. Proposition 2.3(i).

We prove (ii). This time suppose that A is pure, so a is an equivalence; likewise c. Let us show that b is an equivalence. By Example 2.1, we know that $\operatorname{Leff}(X, Y)$ holds, so b is already fully faithful; let us prove that it is essentially surjective. Apply Proposition 2.3(ii), noting that if U is an open neighborhood of Y in X, then $X \setminus U$ is a union of finitely many closed points. The pair $(X, X \setminus U)$ is therefore pure by Proposition 3.3, since at any such point $\mathfrak{p}$ one has $\mathcal{O}_{X,\mathfrak{p}} = A_{\mathfrak{p}}$, which is pure by hypothesis. This proves the assertion.